

ECE433: COMPUTER GRAPHICS

Fall 2025

Project 3:

Visiting a house or a 3D space

(Deadline: Friday, January 16, 2026, 23:55)

General Instructions:

Use the template to fill your report. You can download it atece433_template.zip....

In the theoretical exercises, write in detail your answers in your own words (do not copy-paste). Do write the sources from which you drew them in the “References” and mention them in the document (as in the template).

In the programming ones, present and explain the algorithm you programmed, and present in detail the experimental results of your programs.

In case you downloaded the code from the web (e.g., GitHub, etc.), point out the source in “References” and explain in detail.

Describe your implementation and the machine you worked on (hardware features, OS, compiler, etc.).

Program in C and Linux, use Makefile to compile / link / execute your programs.

Use the Makefile from the lecture examples and always keep the options for MAC in them.

Report your observations and present the results with tables and graphs.

Graphs and figures must be software-generated and not handmade and scanned.

*In general, your text should be well written and legible, while you should FULLY justify the steps you followed, and comment on the results of each exercise. **(20 credits)***

Submit your work via eclass to a zip file (hw3_id1_id2_id3.zip), which will contain the .pdf file with your solutions, results, and comments (LaTeX generated), and your code (.c, .h files, and your Makefile).

It is noted that the groups remain unchanged throughout the semester.

A) Download the examples from the lecture of Friday, December 12, and the mydemo folder (in this project zip) and work the following exercises to get a better understanding of lighting, texturing, and culling procedures.

Exercise 1. (20)

Change the colors of the light, the colors of the sphere, and discuss the results. Enable, disable lighting. Create screenshots for each case and add them to your report, describing your actions.

Exercise 2. (10)

Use different types of culling (front, back, off) and discuss the results. Create screenshots for each case and add them to your report, describing your actions.

Exercise 3. (10)

Use images to apply texture on the cube surfaces. Create screenshots for each case and add them to your report, describing your actions.

B) The purpose of this assignment is the creation of a 3D house and navigation inside it using a moving camera. This project aims at combining the taught material (transformations, modeling, hidden surface removal, lighting, texture mapping) and the corresponding OpenGL and GLUT commands, in order to create a complete three-dimensional animated scene.

The specific house design is taken from the link below, from which you may draw ideas for colors or furniture that you will implement in “logical structures” using OpenGL.

<https://archivaldesigns.com/products/42531-mylitta-modern>

You may choose another design from the same or a different website, provided that you present it in your report and that it satisfies the requirements of the assignment.

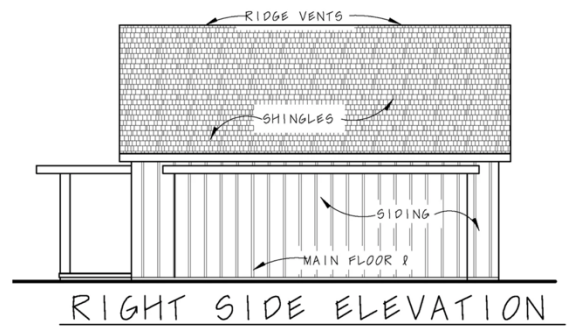
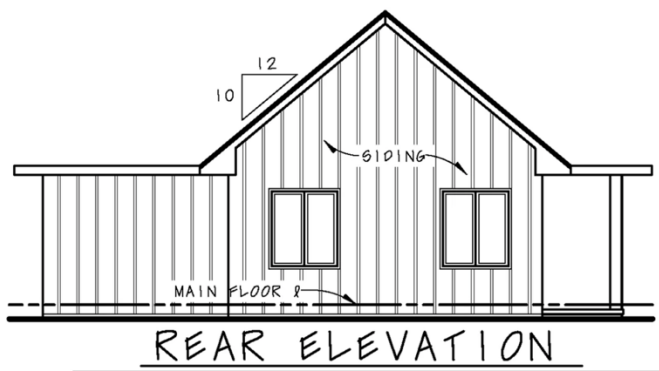
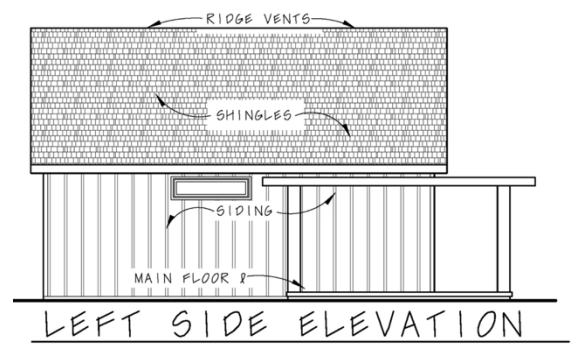


Figure 1. The four sides of the house

1. Construction of the House and Environment (50 points)

Construct the rooms of the house, basic furniture, and the surrounding outdoor area according to **Figure 1**, which depicts a representative floor plan.

- The **walls, floor, and ceiling** must be constructed using the command `glutSolidCube(1.0)` and appropriate modeling transformations (`glScale`, `glTranslate`, `glRotate`).
- Choose dimensions (considering the meter as the unit of measurement) so that they correspond to a real house.
 - Example: bedroom 1 dimensions 3.0 m (length) × 3.0 m (width) × 3 m (height)

- Thickness of exterior walls: 0.25 m
- Thickness of interior walls: 0.10 m
- In Figure 1, room dimensions are given in feet/inches. Use **approximate dimensions in meters** so that the final result resembles the floor plan (multiply feet by 0.3 to convert it to meters).
- You may simplify elements of the design to keep it simple, but sufficient for a house. For example, add the “L” small room in the **BR.1**, or the doors of the closets are closed (no need to show clothes, unless you want to). You may skip the doors to the rooms, but you need to show a door for the main entrance.

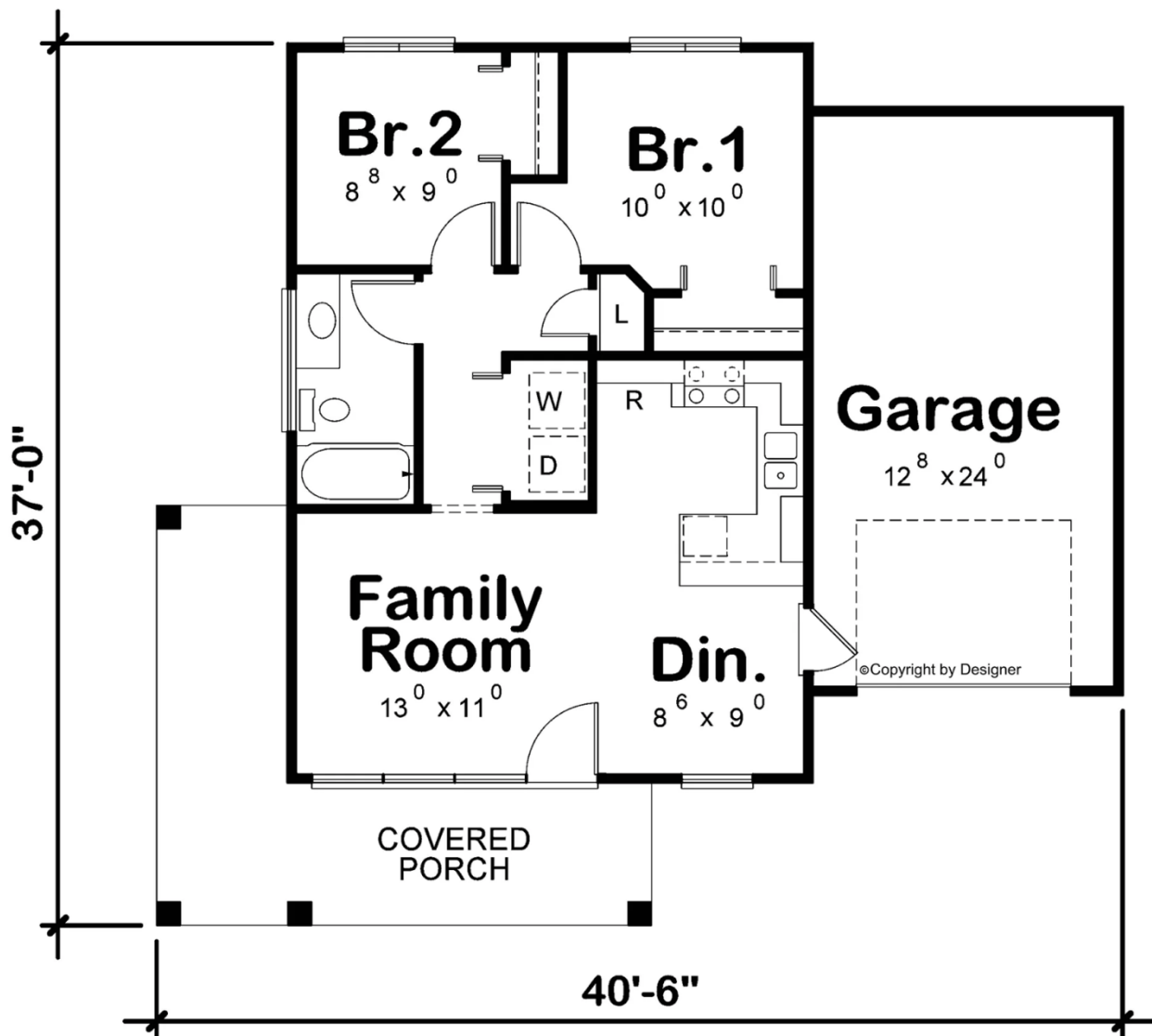


Figure 2. Floor plan of the house.

(50) For the creation of the **outdoor environment**, you must use at least **one parametric surface based on B-spline curves**, including at least one three-dimensional curve.

- You may use multiple curves and height variations to create a more realistic landscape.
- Use the **B-spline functions of OpenGL (GLU library)** for the implementation.

(50) Apply **texture mapping** to at least **two objects**, both in the outdoor environment and inside the house. You may use the images provided in the folder “**texture_images**”, or find others.

Sources for Free Textures:

<https://ambientcg.com>

<https://www.poliigon.com/texture/cracked-aggregate-concrete-texture-brown/8496>

<https://polyhaven.com>

<https://freepbr.com>

<https://3dtextures.me>

<https://www.textures.com>

You may also search for additional sources or create textures from your own images or photographs.

(100) Design **3D furniture** for the **living room–dining area** , and the **bedroom**.

- Furniture may be created using `glutSolidCube(1.0)` or other commands and placed as you see fit.
- Kitchen cabinets are **not required**.
- For doors and windows, simply leave appropriately sized openings.

(50) The **roof** must be constructed using a combination of **two parallelepipeds** (left and right sides) and assigned an appropriate **matte material**.

- Door canopies are **not mandatory**.

2. Moving Camera (100)

Create a **moving camera** that allows the user to navigate through the house using the keyboard:

- **F / f** → move forward
- **B / b** → move backward
- **R / r** → turn right
- **L / l** → turn left
- **U / u** → move up
- **D / d** → move down

The initial camera position must be **outside and in front of the house**.

3. Lighting and Materials (100)

Apply **lighting and material properties** to all objects you design (walls, floor, ceiling, roof, furniture, pathway) that do not have textures.

Lighting must include:

- A **global ambient light**
- A **local light source** that moves together with the camera (either at the camera position or slightly behind it)

Optional Implementations (Bonus to the final grade of the course)

- Transparent glass panes for windows (25 points)
- Front canopy (25 points)
- Back canopy (25 points)
- Two flower pots at the entrance (25 points)
- Sofa and table in the backyard (50 points)

Implementation Details

You must submit **two programs**:

1. **views.c(pp)**

Display different **viewports**:

1. Overall floor plan of the house **without the roof** (orthographic projection)
 2. Overall floor plan **with the roof** (orthographic projection)
 3. Front view of the house (perspective projection)
 4. A suitable interior view of each room, highlighting objects and textures (furniture, floor, etc.)
- **Do not implement lighting or a moving camera** in `views.c(pp)`.

- Implement **textures** and any optional features you wish.
- Include these views in your report.

2. visiting3Dhouse.c(pp)

- Implement **lighting and a moving camera** using perspective projection, as described in the main requirements.
- The rendering window must be in **GameMode (Full Screen)**.

Pressing **Q / q** should terminate both programs.